

Inferact / **MiniMax-M3-EAGLE3**   like 0 Follow  Inferact Inc. 35[Text Generation](#) [Transformers](#) [Safetensors](#) [llama](#) [eagle3](#) [speculative-decoding](#)[draft-model](#) [vllm](#) [torchspec](#) [minimax](#) [text-generation-inference](#) [License: mit](#) Deploy ▾→ Copy to bucket **NEW**

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Model card



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## Model Overview

**Inferact/MiniMax-M3-EAGLE3** is an EAGLE3 draft model for accelerating inference of [MiniMax-M3](#). It is served end-to-end with [vLLM](#) and was trained using [TorchSpec](#) — a torch-native online speculative-decoding training framework that runs FSDP training and vLLM-based target inference concurrently, learning from **MiniMax-M3-regenerated responses and live vLLM-generated hidden states** to match the base model's exact token distribution.

The draft is a **1-layer** dense Llama (LlamaForCausalLMEagle3, ~3.3 B params) operating on MiniMax-M3's hidden\_size=6144 / vocab\_size=200064; at serve time it shares the target's embedding and LM head (EAGLE3). See config.json for the full architecture.

Downloads last month  
**559**



### Safetensors

[Model size](#) 3B params[Tensor type](#) BF16[Files info](#)

### Inference Providers **NEW**

 Text Generation

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### Model tree for Inferact/MiniMax-...

**Base model** [MiniMaxAI/Minimax-M3-pr...](#)**Finetuned**

(1)

[this model](#)

## Performance

All numbers are measured end-to-end against MiniMaxAI/MiniMax-M3-MXFP8 served with vLLM at tensor-parallel-size=4, num\_speculative\_tokens=3, and --enforce-eager. Greedy draft sampling ( topk=1).

| Category  | Dataset                                   | n   | Mean<br>Accept<br>Length | Draft<br>Accept<br>Rate | Per-<br>pos<br>Accep<br>Rate |
|-----------|-------------------------------------------|-----|--------------------------|-------------------------|------------------------------|
| Dialogue  | <a href="#">MT-Bench</a>                  | 80  | 2.698                    | 56.60%                  | 0.749,<br>0.547,<br>0.402    |
| Math      | <a href="#">GSM8K</a>                     | 200 | 3.518                    | 83.93%                  | 0.923,<br>0.839,<br>0.756    |
| Code      | <a href="#">HumanEval</a>                 | 164 | 3.499                    | 83.29%                  | 0.922,<br>0.832,<br>0.744    |
| Math      | <a href="#">MATH500</a>                   | 500 | 3.517                    | 83.90%                  | 0.929,<br>0.841,<br>0.747    |
| Math      | <a href="#">AIME</a>                      | 30  | 3.291                    | 76.36%                  | 0.889,<br>0.763,<br>0.638    |
| Synthetic | speed-<br>bench<br>(16k, low-<br>entropy) | 64  | 2.776                    | 59.21%                  | 0.747,<br>0.576,<br>0.453    |

**Data:** ~456,881 training conversations (the mix2 dataset: SWE-bench-Pro, SWE-bench, OpenCodeInstruct, kimi-mtp), with **all responses regenerated by MiniMax-M3** — preserving the target's reasoning traces and MiniMax-M3 chat formatting.

**Method:** EAGLE3 TTT, ttt\_length=7, max\_seq\_length=32 768, AdamW at  $lr=1 \times 10^{-4}$  (cosine decay to 0, 2 % warmup, max\_grad\_norm=1.0), bf16 + gradient checkpointing, FlexAttention, 1 epoch (~14,277 steps). Trained on **5 × GB300 nodes** (2 nodes FSDP2 draft training, dp=8, global batch 32 + 3 nodes vLLM TP=4 target inference). EAGLE3 aux hidden states from target layers (2, 30, 57) + the final layer. Embedding / LM head / final norm are shared from the target (M3 is a VL model, so these live under the language\_model.\* prefix).

**Core training command** — torchspec.train\_entry spawns the FSDP2 trainer and vLLM inference engines as decoupled Ray actors, streaming hidden states through Mooncake:

```
python3 -m torchspec.train_entry \
  --config configs/vllm_minimax_m3_mix2.yaml
model.draft_model_config=configs/draft_model_config.yaml
training.training_num_nodes=2 \
training.training_num_gpus_per_node=4 \
inference.inference_num_gpus=12 \
inference.inference_num_gpus_per_engine=4 \
inference.vllm.tp_size=4
```

Draft architecture, TTT depth, sequence length, cluster layout, and optimizer are all YAML-configurable — retargeting or scaling is a config

change. See the [TorchSpec repo](#) for full customization instructions.

## Quick Start

### Requirements

- vLLM nightly with MiniMax-M3 support
- Docker image `vllm/vllm-openai:minimax-m3`

### Launch Server (vLLM)

```
vllm serve MiniMaxAI/MiniMax-M3-MXFP8 \
  --tensor-parallel-size 4 \
  --gpu-memory-utilization 0.90 \
  --block-size 128 \
  --speculative-config '{"method": "eagle3",
```